

SELinux the Easy Way

A Practical Guide for QA Engineers

Andrea Manzini

Software Quality Engineer @ SUSE

May 2026





Agenda

1. SELinux Basics and QA Context
2. Modes: Enforcing, Permissive, Disabled
3. Access Decisions: DAC + MAC
4. Policies and Rule Anatomy
5. Labels: Understand, Inspect, Fix
6. Booleans, Ports, and File Contexts
7. QA Testing Guidelines
8. Troubleshooting Denials
9. On RKE2 and Storage providers
10. Live Demos (if time permits)
11. QA Cheat Sheet

What's SELinux?

SELinux (Security-Enhanced Linux) is a security architecture for Linux systems that provides Mandatory Access Control (MAC).

- **MAC vs. DAC:** Unlike traditional Linux permissions (Discretionary Access Control) where users control their files, MAC enforces policies set by the administrator.
- **Default Deny:** If there isn't an explicit rule allowing an action, it is blocked.
- Originally developed by the NSA, now integrated into the mainline Linux kernel.
- Implemented via Linux Security Modules (LSM).

Why SELinux?

- **Containment:** If a service (like a web server) is compromised, SELinux prevents the attacker from accessing the rest of the system or other services.
- **Privilege Escalation Prevention:** Even an exploit running as `root` can be contained if SELinux policies restrict what that specific `root` process can do.
- **Zero-Day Protection:** Helps mitigate the impact of unknown vulnerabilities by enforcing strict behavioral boundaries.
- **Compliance:** Often required by security standards and enterprise environments.

Why QA Cares About SELinux

- SELinux is the **default MAC** on SLES 16, Leap 16, and Tumbleweed
- Customers run SELinux in **Enforcing** mode — so must QA
- SELinux denials look like permission errors — easy to misdiagnose
- A proper bug report with SELinux context saves developers **hours**

"It works on my machine" —
because you had SELinux
disabled.



SELinux Modes



Mode	Behavior	When to use
Enforcing	Blocks + logs denials	Always (production & testing)
Permissive	Logs but does NOT block	Diagnosing SELinux issues only
Disabled	SELinux not initialized	Never (requires reboot + full relabel to re-enable)

```
$ sestatus
```

```
$ getenforce          # Check current mode  
Enforcing
```

```
$ sudo setenforce 0    # Switch to Permissive (temporary)
```

```
$ sudo setenforce 1    # Back to Enforcing
```

```
# Permanent: edit /etc/selinux/config and reboot  
SELINUX=enforcing
```


⚙️ **How SELinux Works** (see link for picture)

SELinux acts as a gatekeeper between a **Subject** (usually a process) and an **Object** (a file, directory, port, etc.).

1. **Subject Request:** A process requests an action (e.g., `read`) on an object (e.g., a file).
2. **DAC Check:** Standard Linux permissions (Owner/Group/Others) are checked first. If this fails, access is denied immediately.

but there's more ...

How SELinux Works

3. **MAC Check:** If DAC passes, SELinux steps in. It looks at the **scontext** (source context — the process) and the **tcontext** (target context — the file).
4. **Policy Evaluation:** The SELinux policy is consulted to see if there is an explicit rule allowing the **scontext** to perform the requested action on the **tcontext**.
5. **Decision:** Allowed if a rule exists, blocked otherwise (Default Deny).

see also [This quick intro](#)

Policies: The Rulebook

A **policy** is the set of rules that defines what each domain (process type) is allowed to do.

Policy types (set in `/etc/selinux/config`):

Policy	Description	Use case
targeted	Only known services are confined, everything else runs unconfined	Default on SLES 16 / Tumbleweed / Leap 16
minimum	Like targeted but fewer services confined	Minimal environments (not shipped on SUSE)
mls	Full Multi-Level Security	Government/military only

Policies are modular — built from independent `.pp` modules:

```
# List loaded policy modules
$ sudo semodule -l | head
abrt
account-utils
accountsdb
acct
afs

# Install a custom module
$ sudo semodule -i my_fix.pp

# Remove a module
$ sudo semodule -r my_fix
```

SLES 16 ships **440+ policy modules** out of the box (based on Fedora upstream policy with SUSE-specific patches). Custom modules are how you extend the policy.
`audit2allow -M` generates them.

```
# See specific module source (if selinux-policy-devel installed)
sudo zypper install selinux-policy-devel
ls /usr/share/selinux/devel/

# Extract a loaded module to CIL (text) format
sudo semodule -E ftpd          # writes ftpd.cil in cwd
sedismod ftpd.pp              # disassemble .pp binary policy

# Or use sesearch to query rules directly
sudo zypper install setools-console
sesearch --allow -s ftpd_t      # all allow rules for ftpd
sesearch --allow -t etc_t      # all rules targeting etc_t
```

Policy Rules: A Concrete Example

A Type Enforcement (TE) rule looks like this:

```
allow  httpd_t  httpd_sys_content_t : file { read getattr open };  
-----  
      source    target                class      permissions  
      domain    type  
└─ "allow this to happen"
```

Read it as: "A process running in the `httpd_t` domain is allowed to `read`, `getattr`, and `open` files labeled `httpd_sys_content_t`."

Policy Rules: A Concrete Example

You can query existing rules with `sesearch` :

```
# What can httpd_t read?
$ sesearch --allow -s httpd_t -t httpd_sys_content_t -c file
allow httpd_t httpd_sys_content_t:file { getattr ioctl lock map open read };

# What can write to files labeled httpd_log_t?
$ sesearch --allow -t httpd_log_t -c file -p write
allow httpd_t httpd_log_t:file { append create write ... };
```

If no rule exists → **denied**. That's the "default deny" principle.

Where Do Policies Live? (on SUSE)

```
/etc/selinux/
├── config                # SELINUX=enforcing, SELINUXTYPE=targeted
└── targeted/
    ├── policy/policy.33  # compiled binary policy (loaded by kernel)
    ├── contexts/         # default contexts for logins, services, etc.
    └── modules/          # installed policy modules (recently migrated here)

/usr/share/selinux/targeted/  # source .pp modules shipped by packages
```


Key packages (zypper):

Package	Contents
<code>selinux-policy</code>	Base policy, file contexts, core types
<code>selinux-policy-targeted</code>	The targeted policy variant (440+ modules)
<code>container-selinux</code>	Policy for Podman/Docker containers
<code>policycoreutils-python-utils</code>	<code>semanage</code> , <code>audit2allow</code> , <code>audit2why</code>

SUSE-specific: policy modules recently migrated from `/var/lib/selinux` to `/etc/selinux` so they're covered by BTRFS snapshots and rollbacks.

What is a Label?

Every object in the system has a **security context** (label).

A label is made of 4 parts: user, role, type, level

```
system_u : system_r : httpd_t : s0:c1,c2
  user      role      type      level
```

Label Fields Explained : User

User (`system_u` , `unconfined_u` , `staff_u`)

- Used by RBAC (Role-Based Access Control) policies.
- [Maps Linux users to SELinux users](#). Controls which roles are available to a user.
- `system_u` = system daemons, `unconfined_u` = regular users (no restrictions)

Label Fields Explained : Role

Role (`system_r` , `object_r` , `unconfined_r`)

- Also used by **RBAC**.
- Bridges users to types — dictates which domains (types) a user can transition into.
- `system_r` = system processes, `object_r` = files/passive objects

Label Fields Explained: Type

Type (`httpd_t` , `user_home_t` , `tmp_t`)

- Used by TE (Type Enforcement) policies.
- **The field you'll work with 95% of the time.** In "targeted" policy, almost all rules check this.
- Processes have a **domain** (`httpd_t`), files have a **type** (`httpd_sys_content_t`).
- Rule format: `allow <domain> <type> : <class> { <permissions> };`

Label Fields Explained: Level

Level (`s0` , `s0:c1,c2`)

- Used by **MLS/MCS (Multi-Level / Multi-Category Security)** policies.
- Defines sensitivity and categories. Crucial for **Container isolation** (e.g., Podman gives each container a unique category like `c1,c2`).

👁️ Viewing Labels

```
# Files
$ ls -Z /var/www/html/index.html
system_u:object_r:httpd_sys_content_t:s0 /var/www/html/index.html

# Processes
$ ps -eZ | grep nginx
system_u:system_r:httpd_t:s0      1234 ?    00:00:00 nginx

# Your own context
$ id -Z
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
```

Rule: if a process in domain `X_t` tries to access a file of type `Y_t`, SELinux checks the policy for an `allow X_t Y_t` rule.

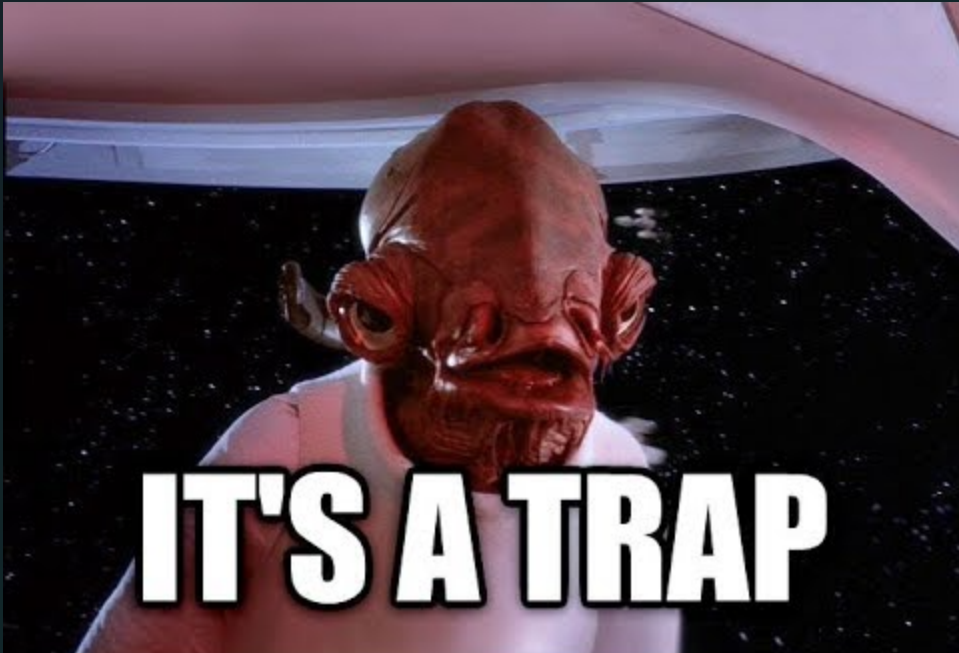
Changing Labels

Tool	Permanent?	Use case
<code>chcon -t httpd_sys_content_t file</code>	No (lost on relabel)	Quick testing
<code>semanage fcontext -a -t httpd_sys_content_t '/srv/web(/.*)?'</code> + <code>restorecon -Rv /srv/web</code>	Yes	Production fix

QA tip: always use `semanage fcontext` + `restorecon` in test procedures. `chcon` changes silently vanish after `restorecon -R` or a full relabel.

⚠ The mv/cp Trap

```
# when a script creates a test config in temp directory  
$ echo "server {}" > /tmp/mysite.conf  
$ ls -Z /tmp/mysite.conf  
unconfined_u:object_r:user_tmp_t:s0 /tmp/mysite.conf
```



⚠ The mv/cp Trap

COPY creates a new file inode → file will be labeled by context rules

```
$ cp /tmp/mysite.conf /etc/nginx/conf.d/  
$ ls -Z /etc/nginx/conf.d/mysite.conf  
unconfined_u:object_r:httpd_config_t:s0      # correct!
```

MOVE keeps original label

```
$ mv /tmp/mysite.conf /etc/nginx/conf.d/  
$ ls -Z /etc/nginx/conf.d/mysite.conf  
unconfined_u:object_r:user_tmp_t:s0          # WRONG! nginx can't read this
```

Fix: `restorecon -v /etc/nginx/conf.d/mysite.conf`

QA Angle: Labels in Bug Reports

When filing a bug where SELinux might be involved, **always include:**

```
# 1. Label of the file/directory that failed
ls -Z /path/to/file

# 2. Label of the process that got denied
ps -eZ | grep <process_name>

# 3. What label the file SHOULD have
restorecon -n -v /path/to/file

# 4. Current SELinux mode
getenforce
```

A bug report that says "permission denied" without this info is **incomplete**.

● Booleans: Policy Switches

Modify policy *behavior* without writing custom rules.

```
# List all booleans related to httpd
$ getsebool -a | grep httpd
httpd_can_network_connect --> off
httpd_can_sendmail --> off
httpd_enable_homedirs --> off

# Allow Apache to make network connections
$ sudo setsebool -P httpd_can_network_connect on
```

The **-P** flag = persistent across reboots. Without it, the change is lost on restart.

Port Labeling

SELinux controls which processes can **bind** to which ports.

```
# See what ports SSH is allowed to use
$ sudo semanage port -l | grep ssh
ssh_port_t                tcp          22

# Move SSH to port 2222 → it will FAIL to start
# Fix: add the port to the policy
$ sudo semanage port -a -t ssh_port_t -p tcp 2222

# Verify
$ sudo semanage port -l | grep ssh
ssh_port_t                tcp          2222, 22
```

Same applies to any service using a non-standard port.

File Context Rules

The **permanent** way to manage labels for custom paths:

```
# Tell SELinux: everything under /srv/myapp is web content
$ sudo semanage fcontext -a -t httpd_sys_content_t '/srv/myapp(/.*)?'

# Apply the rule to existing files
$ sudo restorecon -Rv /srv/myapp
Relabeled /srv/myapp from unconfined_u:object_r:default_t:s0
  to unconfined_u:object_r:httpd_sys_content_t:s0

# Verify
$ ls -Z /srv/myapp/
system_u:object_r:httpd_sys_content_t:s0 index.html
```

✓ QA Angle: Testing with SELinux

Rules for QA test environments:

1. **Always test in Enforcing mode** — matches customer environments
2. Use Permissive **only** to diagnose, then switch back immediately
3. Never put `setenforce 0` in test setup scripts
4. After installing/upgrading a package, run `restorecon -R` on its paths
5. If a test fails with "Permission denied", check SELinux **before** filing a DAC bug

Where to Look for Denials

```
# The audit log (primary source)
$ sudo ausearch -m AVC -ts recent

# Or directly
$ sudo grep "avc: denied" /var/log/audit/audit.log

# Via journald
$ sudo journalctl -t audit --since "10 minutes ago" | grep AVC
```

No denials showing? Check:

- Is `auditd` running? (`systemctl status auditd`)
- Is SELinux actually in Enforcing? (`getenforce`)
- Some denials are "dontaudit" — use `semodule -DB` to disable dontaudit rules and rebuild policy (re-enable with `semodule -B`)

Anatomy of a Denial

```
type=AVC msg=audit(1716000000.123:456): avc:  denied  { read }
    for pid=1234 comm="nginx" name="mysite.conf"
    dev="sda1" ino=5678
    scontext=system_u:system_r:httpd_t:s0
    tcontext=unconfined_u:object_r:user_tmp_t:s0
    tclass=file permissive=0
```

Field	Meaning
<code>{ read }</code>	Which action was denied
<code>comm="nginx"</code>	Which program tried
<code>scontext=...httpd_t...</code>	Process domain (subject)
<code>tcontext=...user_tmp_t...</code>	File type (target)
<code>tclass=file</code>	Object class

⚡ Quick Check: is it a SELinux issue?

```
# Step 1: Something fails. Is SELinux blocking it?  
$ sudo setenforce 0          # Permissive mode  
  
# Step 2: Retry the operation  
$ sudo systemctl restart nginx  
# If it works now → SELinux was blocking it  
  
# Step 3: IMMEDIATELY switch back  
$ sudo setenforce 1  
  
# Step 4: Find the denial  
$ sudo ausearch -m AVC -ts recent
```

Never leave a system in Permissive mode after testing. Automate: set a timer or use `at` to switch back.

audit2why & audit2allow

```
# WHY was it denied?
$ sudo ausearch -m AVC -ts recent | audit2why
Was caused by: Missing type enforcement (TE) allow rule.
Allow rules: allow httpd_t user_tmp_t:file read;

# WHAT rule would fix it?
$ sudo ausearch -m AVC -ts recent | audit2allow
#===== httpd_t =====
allow httpd_t user_tmp_t:file read;

# Generate a loadable policy module
$ sudo ausearch -m AVC -ts recent | audit2allow -M my_nginx_fix
$ sudo semodule -i my_nginx_fix.pp
```

Warning: `audit2allow` may suggest overly broad rules. Always check what it proposes before loading.

setroubleshoot / sealert

Human-readable analysis of denials:

```
$ sudo sealert -a /var/log/audit/audit.log

SELinux is preventing nginx from read access on the file mysite.conf.

***** Plugin restorecon (99.5 confidence) suggests *****
If you want to fix the label:
    restorecon -v /etc/nginx/conf.d/mysite.conf

***** Plugin catchall (1.49 confidence) suggests *****
If you want to allow httpd_t to read user_tmp_t files:
    audit2allow -M mypol
```

sealert ranks suggestions by confidence — start from the top.

Container Volume Isolation (MCS)

SELinux uses **MCS** (Multi-Category Security) to isolate containers.

- Each container gets unique category pair: `s0:c123,c456`
- Prevents containers from accessing each other's volumes
- Even if running as same UID

Example Pod security context:

```
spec:
  securityContext:
    seLinuxOptions:
      level: "s0:c0.c1023" # allow broad category range
```

Kubernetes 1.27+: Efficient mount-time labeling — no recursive `chcon` on large volumes. See [K8s efficient relabeling blog](#).

Storage Backend Issues & Fixes

Backend	Issue	Fix
Longhorn	iSCSI <code>dac_override</code> denials (container-selinux v2.189.0+)	Load custom module: <code>echo '(allow iscsid_t self (capability (dac_override)))' > fix.cil && semodule -vi fix.cil</code> KB
local-path-provisioner	SELinux blocks hostPath access	<code>chcon -t container_file_t -R /opt/local-path-provisioner</code> Issue
NFS volumes	Container can't mount NFS	<code>setsebool -P virt_use_nfs on</code>

RKE2 + SELinux References

Official docs:

- [RKE2 SELinux Documentation](#)
- [SUSE RKE2 SELinux Guide](#)
- [Kubernetes Security Contexts](#)

Known issues:

- [RKE2 Known Issues](#) — SELinux module upgrade problems
- [Downstream cluster SELinux setup](#)

QA testing notes:

- Always test with `selinux: true` in RKE2 config
- Check volume mount denials: `ausearch -m AVC | grep mount`

Demo Setup

Prerequisites: openSUSE Tumbleweed with `distrobox` and `podman` installed.

```
# From this repo's directory:
$ ./selinux-lab.sh

# This will:
# 1. Create a Tumbleweed distrobox named "selinux-lab"
# 2. Install SELinux tools, nginx, audit utilities
# 3. Copy a sample audit log for offline demos

# Enter the lab:
$ distrobox enter selinux-lab
```

All demo commands in the following slides can be run inside this container. Note: actual enforcement requires a system with SELinux enabled in the kernel (Tumbleweed, SLES 16).

★ Demo: nginx Can't Read Its Config

```
# 1. Create a config file in /tmp
echo 'server { listen 8080; root /usr/share/nginx/html; }' \
  > /tmp/demo.conf

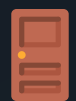
# 2. Move it (not copy!) to nginx config dir
sudo mv /tmp/demo.conf /etc/nginx/conf.d/

# 3. Check the label – it's WRONG
ls -Z /etc/nginx/conf.d/demo.conf
# → user_tmp_t (should be httpd_config_t)

# 4. On a real enforcing system: nginx would fail to start
#     Check the audit log for the AVC denial

# 5. Fix it
sudo restorecon -v /etc/nginx/conf.d/demo.conf
# → Relabeled to httpd_config_t

# 6. Verify
ls -Z /etc/nginx/conf.d/demo.conf
```



Demo: Service on a Non-standard Port

```
# 1. Check which ports httpd_t can bind to
sudo semanage port -l | grep http_port
# → http_port_t      tcp      80, 81, 443, 488, ...

# 2. Try to use port 8888 → SELinux would block it
#     The AVC denial would show:
#     denied { name_bind } for ... scontext=...httpd_t...

# 3. Add port 8888 to the allowed list
sudo semanage port -a -t http_port_t -p tcp 8888

# 4. Verify
sudo semanage port -l | grep http_port
# → http_port_t      tcp      8888, 80, 81, 443, ...

# 5. Now the service can bind to port 8888
```

QA Cheat Sheet

Task	Command
Check SELinux mode	<code>getenforce</code>
View file labels	<code>ls -Z /path/to/file</code>
View process labels	<code>ps -eZ grep <name></code>
Find recent denials	<code>ausearch -m AVC -ts recent</code>
Explain a denial	<code>ausearch -m AVC -ts recent audit2why</code>
Fix a wrong label	<code>restorecon -v /path/to/file</code>
Check expected label	<code>restorecon -n -v /path/to/file</code>
Add permanent label rule	<code>semanage fcontext -a -t <type> '/path(/.*)"'</code>
Toggle boolean	<code>setsebool -P <bool> on</code>

🙏 Thank You!

Questions?

Andrea Manzini

Software Quality Engineer @ SUSE

Lab setup & slides:

https://github.com/ilmanzo/suse_presentations

